

These functions either generate or verify a signature of type **Signature** defined by the following mapping.

Mapping (Version 1.0)

```
struct {
    byte[CRYPTO_GROUP_SIZE_BYTES] r,
    byte[CRYPTO_GROUP_SIZE_BYTES] s
} Signature
```

3.5.3.1. Signature

Function and description

```
Signature
Crypto_Sign(
    PrivateKey privateKey,
    byte[] message)
```

Returns the signature of the **message** using the **privateKey**.

Mapping (Version 1.0)

```
Crypto_Sign(privateKey, message) :=
    Signature ECDSASign(dU := privateKey, M := message)
```

ECDSASign() SHALL be the ECDSA signature function as defined in Section 4.1 of [SEC 1](#) using **Crypto_Hash()** as the underlying hash **Hash()** function.

3.5.3.2. Signature verification

Function and description

```
boolean
Crypto_Verify(
    PublicKey publicKey,
    byte[] message,
    Signature signature)
```

Verifies the **signature** of the **message** using the **publicKey**, returns **TRUE** if the verification succeeds, **FALSE** otherwise.